

First Detection:

Toward the Study of AI's Impact on Legal
Drafting

AECA Granada 2026

Pilot Project

Project's team

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AI is transforming society faster than it can be regulated.

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The judicial system is one clear example.

Judicial institutions are adopting AI tools.

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Studying AI's impact on the judicial system is urgent.

Research Gap

Researchers can't test how these tools are implemented

⇒ **Regulation lacks empirical evidence**

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We cannot systematically detect AI use in rulings

⇒ **AI use cannot be held to account**

Can the use of AI be reliably detected in judicial rulings?

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Do existing Gen-AI text detection approaches generalize to the legal domain?

Do these metrics work...

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in a controlled, synthetic benchmark?

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on 45 years of real rulings?

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in a controlled, synthetic benchmark?

on 45 years of real rulings?

on real documents with hallucinations?

3 metrics

×

3 model pairs

×

3 datasets

Analysis 1

11,570 real rulings, generated five different ways.

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Rewrite Trim Translate back Reverse ruling Partial
rewrite

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Generated with llama3.1-8b-instruct

57,850

synthetic documents generated

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Metric scoring: still in progress.

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Infrastructure, not method, is the bottleneck.

28,742 documents
all Spanish TC rulings, 1980–2026

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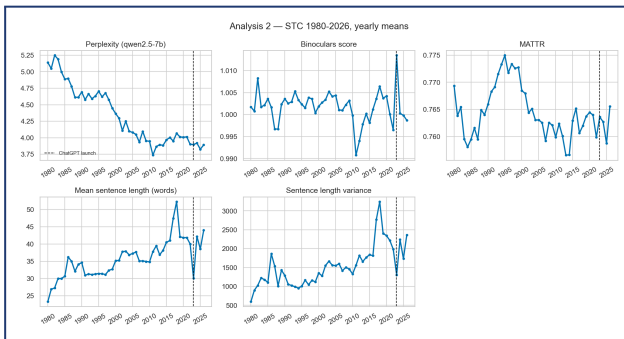
Scored independently by two model pairs.

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Reference point: ChatGPT's public launch, November 2022.

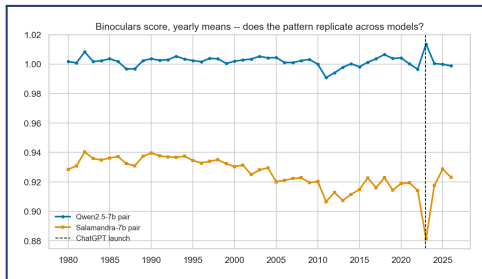
Results A2 – Long-run trends



Yearly means, Qwen2.5-7B pair, 1980–2026

Strong 45-year trends – no break at the ChatGPT launch.

Results A2 – Cross-model replication



Binoculars score, yearly means – Qwen2.5-7B vs. Salamandra-7B

Same documents, same year – opposite direction.

1,518

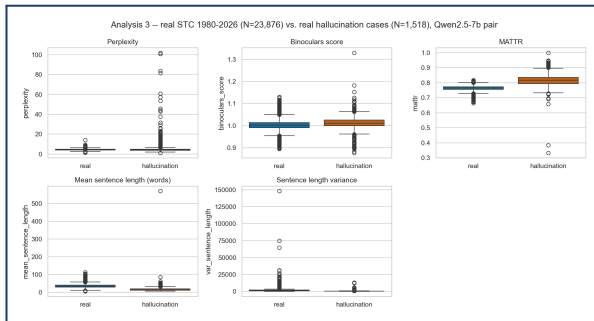
real hallucination incidents, worldwide

1,518

real hallucination incidents, worldwide

Caveat: we score the court's own response, not the hallucinated filing.

Results A3 – Real vs. hallucination cases



N = 23,876 real, N = 1,518 hallucination – Qwen2.5-7B pair

Real and hallucinated text score nearly identically.

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Next: a validated, strategy-stratified benchmark.

Limitations – Benchmark construction

*Similar enough to be credible, different enough
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trivially detectable

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Partial rewrite

hard, policy-relevant case

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Report results per strategy – not pooled.

Limitations – Analyses 2 & 3

No detector found a clean break at the ChatGPT era –
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Hallucinated and real legal text score the same on Binoculars.

Where this leaves us

We cannot yet confirm that these metrics detect AI use in Spanish TC rulings.

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Genuinely low AI use *or* a drafting process already low-perplexity by design?

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We can't adjudicate yet – the benchmark isn't ready.

Validating a comparable benchmark

Register / template conformance

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Blinded human-expert rating

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Legal coherence checks

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Legal coherence checks

Dose-response validation

Future directions

Other metrics: DetectGPT / Fast-DetectGPT

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Cross-family testing

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Ensemble across model pairs

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Classical stylometry

Thank you